

# Customer Support Ticket Analyzer

## 1. Project Overview

The Customer Support Ticket Analyzer is a Python-based project designed to manage and analyze customer support tickets. The project allows users to add ticket details, validate ticket priorities, and generate useful insights from customer issues.

This project uses Python concepts such as lists, dictionaries, loops, conditional statements, functions, and string manipulation to organize and analyze ticket data.

## 2. Project Objectives

- To create and manage customer support tickets.
- To collect customer names and issue descriptions.
- To validate ticket priorities (High, Medium, Low).
- To automatically assign unique ticket numbers.
- To count tickets based on priority.
- To identify common keywords in customer issues.
- To analyze ticket descriptions and generate useful insights.
- To practice Python programming and data analysis concepts.

## 3. Technologies Used

- **Programming Language:** Python
- **Development Environment:** Jupyter Notebook
- **Libraries:** Python built-in functions and data structures

## 4. Project Features

### 4.1 Ticket Management

- Accepts the number of tickets that the user wants to add.
- Collects customer names, issue descriptions, and priority levels.
- Assigns a unique ticket number to each ticket.
- Stores ticket details in a dictionary using lists.

## 4.2 Priority Validation

- Supports three priority levels: High, Medium, and Low.
- Validates user input to ensure only accepted priority values are entered.
- Prompts the user again when an invalid priority is provided.

## 4.3 Ticket Analysis

- Counts the total number of tickets.
- Calculates the number of High, Medium, and Low priority tickets.
- Identifies the ticket with the longest issue description.
- Provides a summary of the collected ticket information.

## 4.4 Keyword-Based Issue Insights

- Searches ticket descriptions for selected keywords.
- Counts how many ticket descriptions contain words such as:
  - Poor
  - Good
  - Slow
  - Excellent
- Helps identify recurring issues and positive feedback.

## 4.5 Unique Word Analysis

- Extracts unique words from customer issue descriptions.
- Identifies the total number of unique words.
- Displays the sorted list of unique words.
- Helps understand the different terms used in customer feedback.

# 5. Python Concepts Used

| Concept      | Purpose                                                 |
|--------------|---------------------------------------------------------|
| Variables    | Store ticket information and analysis results           |
| Lists        | Store customer names, ticket numbers, and issue details |
| Dictionaries | Organize ticket data                                    |

|                        |                                          |
|------------------------|------------------------------------------|
| For Loops              | Iterate through tickets and descriptions |
| While Loops            | Validate priority inputs                 |
| Conditional Statements | Check priorities and keywords            |
| Functions              | Perform reusable analysis operations     |
| String Methods         | Process and analyze issue descriptions   |
| Sets                   | Identify unique words                    |
| Input and Output       | Collect and display ticket information   |

## 6. Project Workflow

1. Start the program.
2. Ask the user how many tickets they want to add.
3. Collect customer details for each ticket.
4. Validate the priority of each ticket.
5. Store the ticket information in the data structure.
6. Display successful ticket addition messages.
7. Analyze ticket priorities and issue descriptions.
8. Count selected keywords in customer feedback.
9. Identify unique words in the issue descriptions.
10. Display the final analysis results.

## 7. Analysis Performed

### 7.1 Ticket Priority Analysis

The project calculates the number of tickets belonging to each priority level. This helps understand how customer issues are distributed across different urgency levels.

### 7.2 Keyword-Based Analysis

The project counts selected keywords in ticket descriptions to identify frequently mentioned issues and positive feedback.

### 7.3 Issue Description Analysis

The project identifies the ticket with the longest issue description and calculates the number of unique words used across ticket descriptions.

## 8. Sample Output

High-priority Tickets: 5

Medium-priority Tickets: 4

Low-priority Tickets: 3

Keyword-Based Issue Insights

Tickets containing 'poor': 3

Tickets containing 'good': 2

Tickets containing 'slow': 1

Tickets containing 'excellent': 1

Unique Words Analysis

Count of Unique Words: 32

Sorted Unique Word List: [...]

**Note:** The numbers above are illustrative. Actual results depend on the ticket details entered by the user.

## 9. Project Outcomes

- Successfully created a basic customer support ticket management system.
- Organized customer support data using Python data structures.
- Implemented input validation for ticket priorities.
- Performed basic analysis of ticket priorities and issue descriptions.
- Extracted keyword counts and unique words from customer feedback.
- Practiced fundamental Python programming and data analysis techniques.

## 10. Conclusion

The Customer Support Ticket Analyzer demonstrates how Python can be used to manage and analyze customer support information. The project provides a basic ticket management system and generates

simple insights from customer feedback using loops, conditional statements, lists, dictionaries, functions, and string operations.

This project helped develop practical skills in Python programming, data organization, data cleaning concepts, and basic text analysis. It can be further improved by integrating a CSV dataset, adding visualizations, and developing an interactive dashboard.

## **11. Future Enhancements**

- Store ticket information in CSV or database files.
- Add the ability to search and update existing tickets.
- Generate graphical reports using Matplotlib.
- Perform more advanced text analysis using Python libraries.
- Build an interactive dashboard to visualize ticket trends.
- Add automated ticket categorization and sentiment analysis.
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